

HAKIM AREZKI, PhD

AI & GENERATIVE AI ARCHITECT — LLM, RAG AND AGENTS

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AI architect and technical lead with 12 years of experience (15 including doctoral research) in Data and Artificial Intelligence, including 8 years in regulated financial services (BIL, BPCE) and large industrial groups (TotalEnergies, RATP). PhD from CentraleSupélec.

Design and operation of GENIA, a secure enterprise Data and AI platform in a banking environment: Generative AI, LLM and agent capabilities delivered as governed products, enabling business domains to build and run AI use cases at scale.

Hands-on across the full scope: multi-LLM orchestration with routing, caching, fallback and latency control; RAG and Document AI pipelines; AI agents; LLMOps and MLOps industrialisation; API and event-driven integration; AI governance covering access policies, model selection, hallucination risk, confidentiality and auditability.

Strong data platform foundation (Snowflake, Databricks, dbt, Power BI) ensuring AI capabilities are fed by governed, trusted and secured enterprise data rather than isolated pipelines.

AI delivery in regulated environments

BIL, Banque Internationale à Luxembourg (since 2026): enterprise Data and AI platform, multi-LLM mesh, RAG and Document AI, AI agents, LLMOps, AI governance, RBAC and Row-Level Security, GDPR, DORA.

TotalEnergies (2020-2025): group-scale industrialisation of Machine Learning and Generative AI use cases, AI agents, Document AI, RAG pipelines, MLOps and LLMOps.

RATP, MSM programme (2018-2020): predictive models, computer vision and OCR, real-time architectures for multimodal mobility supervision.

BPCE (2017-2018): AI-driven financial crime detection — money laundering, terrorism financing, sanctions and embargo screening, fraud — enterprise catalog and lineage, GDPR compliance.

Core skills

AI and agentic platform architecture

Architecture of agentic AI platforms: orchestration, tool calling, memory, guardrails · Multi-LLM orchestration, model routing, caching, fallback and latency control · RAG, hybrid search, reranking and Document AI pipelines · Integration with the enterprise data layer and business applications · Connectors, APIs and event-driven integration patterns

AI lifecycle industrialisation

Deployment and operation of frontier and open-source models · Model lifecycle: deploy, monitor, retrain · Versioning, CI/CD for models and prompts · Evaluation frameworks: RAGAS, Hit Rate, MRR, Recall@k, nDCG · Performance, usage, cost and reliability monitoring · Production deployment at scale

Machine Learning and Data Science

Predictive and classification models · Natural language processing, computer vision and OCR · Time series and forecasting · Anomaly detection · Recommendation engines · Industrialisation and scaling of model code

AI governance, risk and compliance

Governance frameworks: access policies, model selection and approval, usage rules · Risk management: hallucination control, guardrails, human-in-the-loop, model risk · Data confidentiality and regulatory constraints: GDPR, DORA, EU AI Act awareness · Security and IAM for AI usage · Auditability and traceability of AI outputs

Enablement and adoption

Catalog of reusable components: agents, prompts, APIs, templates · Developer and user experience · Onboarding of business domains and promotion of adoption · Training of business and technical teams

Technical leadership

Leadership of multi-disciplinary platform teams of around 20 people · Architecture governance, structuring decisions and technical standards · Dialogue with business and executive sponsors · Balance of innovation, cost and risk · Agile delivery: Scrum, Kanban, SAFe, Jira, Confluence

Technology environment

AI and Generative AI: OpenAI, Azure OpenAI, Microsoft Copilot et Copilot Studio, Claude, Mistral, LLaMA, Hugging Face, RAG, AI agents, Document AI · Frameworks: LangChain, LlamaIndex, Semantic Kernel, MLflow, Dataiku, PyTorch, TensorFlow, scikit-learn · RAG and vector search: FAISS, PGVector, BM25, hybrid search, reranking, RAGAS · Document AI: Azure Document Intelligence, vision-language models, Docling, Tesseract, OpenCV · MLOps and observability: MLflow, OpenTelemetry, Prometheus, Grafana, Langfuse, Arize · Data and Cloud platforms: Snowflake, Databricks, dbt, Azure, GCP, AWS · Data Engineering: Python, SQL, Spark, PySpark, Kafka, Airflow · APIs and integration: FastAPI, REST APIs, Redis Streams, event-driven architectures · Infrastructure: Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, GitLab CI/CD

Professional experience

Data & AI Programme Lead — BIL, Banque Internationale à Luxembourg

January 2026 to present

Leading the design, industrialisation and operation of GENIA, the bank's enterprise Data & AI platform, delivering governed, shared Generative AI capabilities to business domains in a regulated banking environment.

Contributed to the platform vision, target architecture and operating model, treating AI capabilities as reusable products rather than isolated use cases.

Translated business priorities into a platform roadmap and a prioritised portfolio of shared capabilities, agreed with business and IT stakeholders.

Defined platform steering indicators covering usage, performance, adoption and cost of AI services (FinOps).

Drove FinOps cost control and optimisation of model and API consumption, including model selection, routing and caching strategies.

Led the modernisation of the legacy IBM Cognos decision-support estate towards Snowflake and Power BI: inventory and rationalisation of the reporting portfolio, redesign of data models, semantic layer rebuild, reconciliation, performance testing and user enablement.

Contributed to the AI governance framework: access policies, model selection and approval, usage rules, alignment with the bank's Data and Digital governance bodies.

Addressed AI-specific risks: hallucination control, guardrails, human-in-the-loop validation and evaluation gates before go-live.

Ensured compliance with confidentiality and regulatory constraints: RBAC, Row-Level Security, GDPR, encryption, traceability and auditability, in line with banking requirements including DORA.

Coordinated Data, Engineering, Architecture, AI, DevOps and business stakeholders, and managed technology partners.

Lead Data & AI Platform — TotalEnergies

December 2020 to December 2025

Group-scale industrialisation of Machine Learning and Generative AI use cases on an enterprise Data Platform serving multiple business domains: retail, human resources and industrial operations.

Led multi-disciplinary teams of around 20 people across business, Data Engineering, Data Science, architecture, cloud and DevOps.

Managed external partners and service providers contributing to platform and use-case delivery.

Moved use cases from experimentation to production with the associated MLOps and LLMOps practices: CI/CD, monitoring, evaluation, observability and continuous improvement.

Contributed to the design and evolution of the enterprise Data Platform on Snowflake, supporting multiple business domains and large heterogeneous volumes.

Defined data architecture and technical standards covering data modelling, MDM, APIs, governance, security and platform integration.

Translated business requirements into reusable data products and platform capabilities.

Contributed to Master Data Management initiatives and implemented data quality, governance, lineage and access-control practices.

Supported the adoption and continuous evolution of AI services, data products and BI capabilities across the organisation.

Head of Data — RATP, MSM programme with Alstom, Renault and IRT SystemX

August 2018 to December 2020

Leadership of the Data and AI workstreams of a multi-partner multimodal mobility programme bringing together several industrial partners and a technology research institute.

Led AI and data initiatives across the programme, from use-case framing through architecture design to production deployment.

Managed internal and external teams — AI engineers, data scientists, developers: scope definition, resource planning, roadmap alignment, technical governance and delivery standards.

Designed microservices-based architectures for real-time supervision and incident management.

Built and deployed a multi-layered Data Lake on Azure for scalable ingestion, processing and analytics.

Developed predictive models for passenger flow, incident duration and operational planning.

Implemented anomaly detection and log analysis to ensure data quality and regulatory compliance.

Trained business and technical teams on AI platforms, data visualisation and process automation.

Delivered functional demonstrators for smart mobility supervision at La Défense.

Lead Data Science & Governance — BPCE, Banking & Insurance

January 2017 to August 2018

Data and AI programme within a major French banking and insurance group, covering anti-money-laundering, counter-terrorism financing, sanctions and embargo screening, fraud detection and GDPR compliance of financial data processing.

Led the build of an end-to-end AI-driven fraud detection engine and the automation of alert, sanctions and embargo workflows.

Designed and deployed the enterprise data dictionary and lineage capability on Apache Atlas, establishing traceability of financial data assets from source systems through to regulatory reporting.

Structured metadata, ownership and data definitions across compliance, sanctions and fraud domains.

Ensured GDPR compliance of financial data processing chains: access control, traceability and data protection.

Defined and tracked detection-system performance indicators as service-level metrics, with Power BI reporting for risk and compliance teams.

Ran Agile delivery adapted to data-centric work in a banking environment: kickoff, steering committees, retrospectives, testing and go-live reviews.

Lead Data Scientist — Luxury Retail

September 2015 to January 2017

Built a customer intelligence engine: profile classification, behaviour prediction, churn and retention modelling, and a recommendation engine.

Defined and tracked retail performance indicators, with real-time reporting to business teams.

Refactored and industrialised model code for scalability and deployment.

Data Scientist — Beamlabs / Française des Jeux

November 2014 to September 2015

Built a Big Data foundation for real-time marketing: distributed architecture, streaming ingestion, indexing.

Developed a real-time recommendation engine and predictive marketing models.

R&D Engineer, PhD — CentraleSupélec and École Polytechnique

February 2010 to October 2014

Applied Machine Learning to photovoltaic cell performance optimisation and transistor modelling; market analysis of photovoltaic expansion in Europe.

Developed simulation and data analysis algorithms, and scientific data visualisation.

Education

PhD, CentraleSupélec, Université Paris-Saclay, Orsay (2011-2014)

Master of Sciences, Université Pierre et Marie Curie, Paris VI (2010-2011)

Master's degree in Engineering Sciences, Université Paris Diderot, Paris VII (2009-2010)

Languages

French: native

English: fluent professional proficiency

German: notions

Publications and speaking

Architecture of a Public Transport Supervision System Using Hybridization Models Based on Real and Predictive Data, IEEE / DSD 2020, pp. 440-446

Disruption Impact on the Offer and Demand on the Public Transport Network in Paris

Big Data Paris 2018, speaker: "We can tell you by what external context your sales are influenced"

Training delivered

Advanced Python for Machine Learning (SQLI Institute, 2019) · Datalab (Saagie, 2019) · SQL for Data Science (Fitec, 2018) · ELK (Fitec, 2017) · Kafka (Air4Data / SQLI Institute, 2017) · DataViz, Tableau and Power BI (SQLI Institute, 2017)